

# WELL COMPLETION REPORT (SYNTHETIC)

## OIL-DGB-007 - Digboi-007

NWIS prototype document - not an Oil India Limited operational record

|                         |                                     |
|-------------------------|-------------------------------------|
| API Number              | OIL-ASM-000073                      |
| Field / District        | Digboi / Tinsukia                   |
| Block                   | PEL-ASSAM-01                        |
| Basin                   | Assam-Arakan                        |
| Formation               | Barail (BAR)                        |
| Spud / Completion       | 2013-10-02 / 2014-02-12             |
| Measured / TVD          | 2739 m / 2705 m                     |
| Trajectory              | Vertical; inc 4.3 deg; az 118.1 deg |
| Formation Interval      | 1249-1928 m                         |
| Lithology               | Interbedded sandstone and shale     |
| Pore Pressure Reference | 2434 psi                            |
| Fracture Gradient       | 15.60 ppg                           |
| Mud System              | PHPA Water-Based Mud                |
| NWIS Well Risk          | Moderate (49.2/100)                 |

## Casing Program

| String            | Set Depth (m) | Purpose      |
|-------------------|---------------|--------------|
| 20 in conductor   | 80            | Conductor    |
| 13 3/8 in casing  | 684           | Surface      |
| 9 5/8 in casing   | 1588          | Intermediate |
| 7 in liner/casing | 2410          | Production   |

## Significant Historical Events and Lessons

| Date / Depth         | Event          | Severity | Evidence / mitigation / lesson                                                                                                                                                                                             |
|----------------------|----------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2013-10-19<br>482 m  | Kick           | Critical | Flow-out exceeded flow-in and gas units increased. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Treat pit gain plus gas increase as an early influx indicator.          |
| 2013-12-17<br>1672 m | Kick           | Critical | Flow-out exceeded flow-in and gas units increased. Mitigation: Maintain trip margin and use conservative connection practices. Lesson: Escalate connection-gas trends before entering correlated pressure zone.            |
| 2013-11-27<br>1285 m | Torque Spike   | High     | Sharp torque excursion occurred with RPM instability. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Torque-rate change is a useful precursor even when pressure is stable.            |
| 2013-11-29<br>1319 m | Torque Spike   | High     | Rotary speed fluctuated during constant pump rate. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Torque-rate change is a useful precursor even when pressure is stable.                          |
| 2014-01-16<br>2243 m | NPT            | High     | No depth progress occurred during the recorded interval. Mitigation: Verify equipment health before resuming drilling. Lesson: Consistent NPT coding enables meaningful cross-well comparison.                             |
| 2013-10-17<br>445 m  | Pressure Spike | Medium   | Circulating pressure climbed without equivalent torque change. Mitigation: Use pressure trend thresholds referenced to depth and flow rate. Lesson: Rate-of-change in pressure is more informative than a fixed threshold. |
| 2013-10-24<br>592 m  | Kick           | Medium   | Flow-out exceeded flow-in and gas units increased. Mitigation: Re-establish overbalance and monitor pit/flow balance. Lesson: Maintain mud-weight window using offset pressure evidence.                                   |

## Completion Summary

The synthetic well reached 2739 m MD in the Barail interval. The NWIS knowledge record links 14 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.